

KnowLiq

An Adaptive AI Learning Platform

1. WHAT IS KNOWLIQ?

KnowLiq is an AI-powered learning platform designed for students.

Most AI tools can answer questions, summarize documents, or explain topics. KnowLiq goes a step further: it keeps track of **what a student needs to learn, what they have already studied, and how well they are retaining it.**

Its goal is to turn AI from a simple question-answering tool into a **personal learning system.**

2. HOW KNOWLIQ WORKS

Step 1 — Upload Past Papers

The student uploads their past examination papers. KnowLiq uses AI to analyze these papers and identify:

- Topics and subtopics
- How frequently topics appear
- Their relative importance
- Total marks associated with them

This creates a structured syllabus for the subject.

Step 2 — Build a Personalized Learning Plan

KnowLiq creates a learning workspace for the student. It tracks each subtopic using information such as:

- Historical performance & retention status
- Topic importance & marks distribution
- When the topic was last studied

The system uses this information to decide which concepts should receive attention next.

Step 3 — AI Tutor

The student starts learning with the AI tutor. Instead of giving large amounts of information at once, KnowLiq teaches **one concept at a time.**

The tutor can adapt its teaching style, including:

- Visual explanations and analogies
- Interactive questioning
- Concise vs. detailed explanations

It also remembers relevant parts of previous conversations so learning can continue seamlessly across sessions.

Step 4 — Continuous Assessment

After the student has learned enough material, KnowLiq generates a diagnostic test based on the concepts they studied. The AI evaluates the student's answers and determines how well each subtopic was understood.

Step 5 — Update and Adapt

The evaluation updates the student's learning state. Each subtopic receives a retention status:

Green

Strong retention and high comprehension.

Yellow

Needs revision to reinforce key concepts.

Red

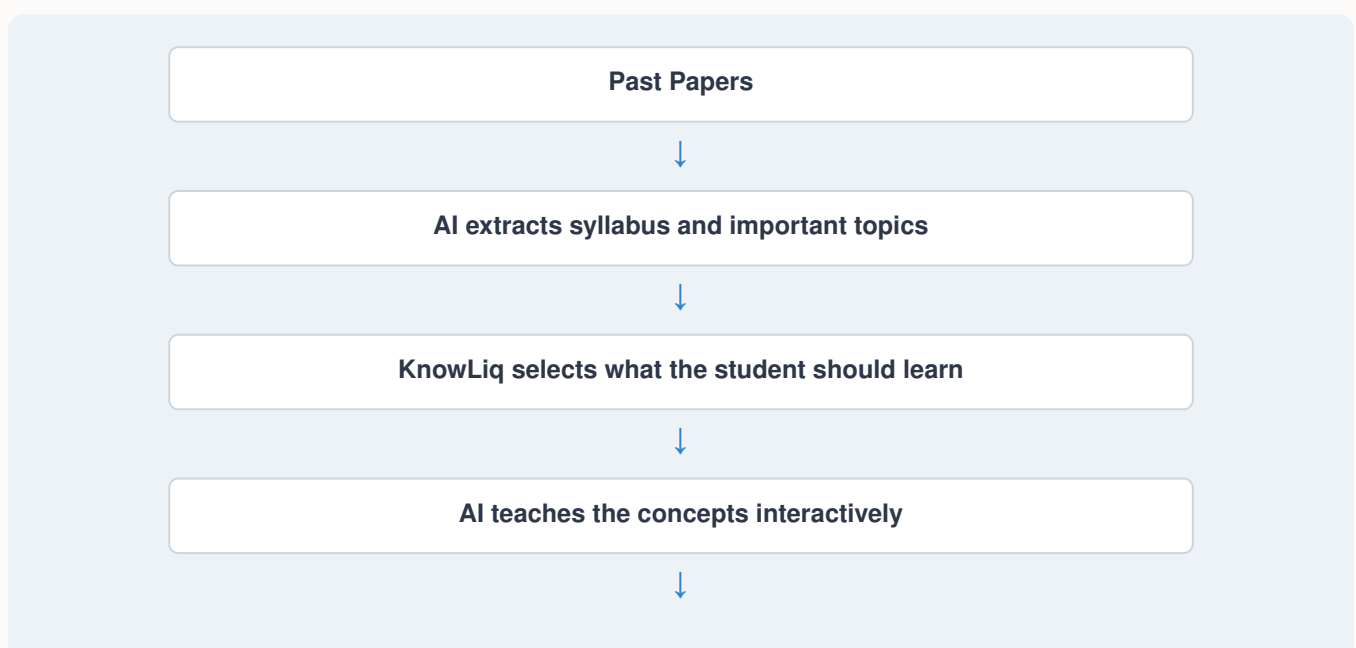
Requires attention and immediate targeted re-teaching.

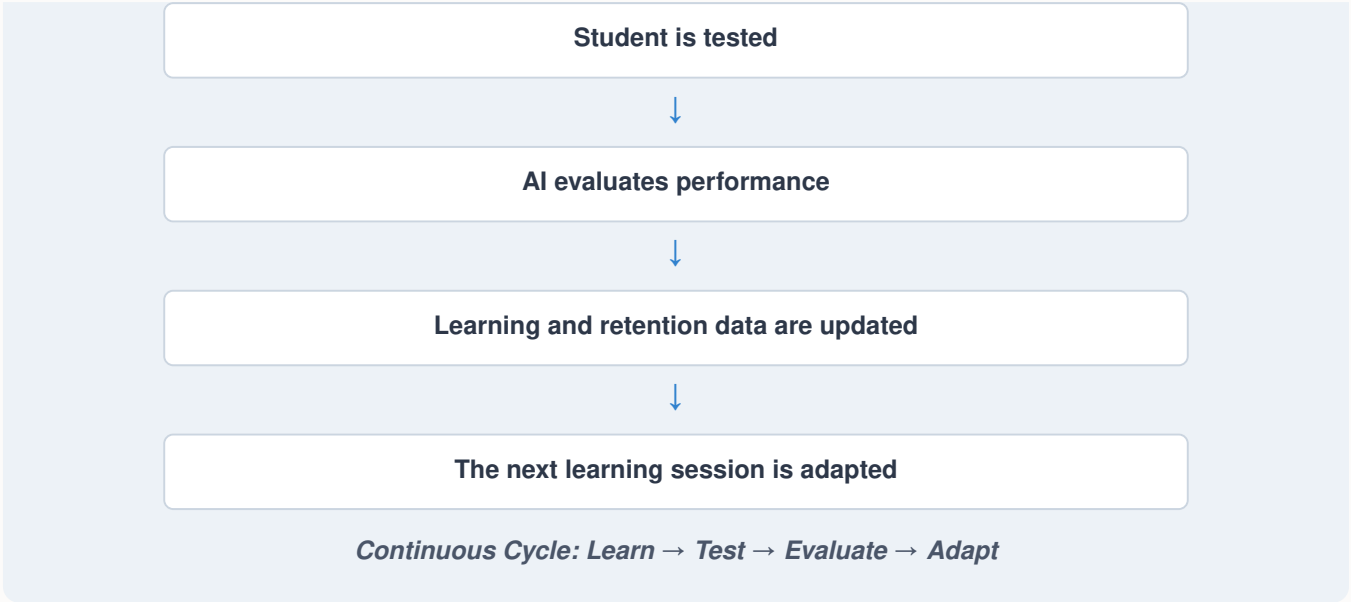
Untested

Not yet evaluated in diagnostic testing.

This information continuously influences what KnowLiq recommends next.

3. THE COMPLETE LEARNING LOOP





4. WHAT WE BUILT

We built the core backend system that powers this process, including:

- AI-powered past-paper analysis
- Automatic syllabus and topic extraction
- Adaptive topic selection
- Personalized AI tutoring
- Short-term and long-term conversation memory
- Semantic retrieval of relevant previous knowledge
- Automatic diagnostic test generation
- AI-based answer evaluation
- Topic performance and retention tracking
- Persistent student learning state

5. WHY KNOWLIQ IS DIFFERENT

KnowLiq is not designed to be just another AI chatbot.

Normal AI Chatbot	KnowLiq System
Focuses primarily on answering: “What is this?”	Designed to dynamically answer: “What should I learn, how should I learn it, did I understand it, and what should I study next?”

The result is an AI system that focuses not only on **providing information**, but on managing the **learning process itself**.

Core Concept: KnowLiq transforms academic material into a continuously adapting learning journey.